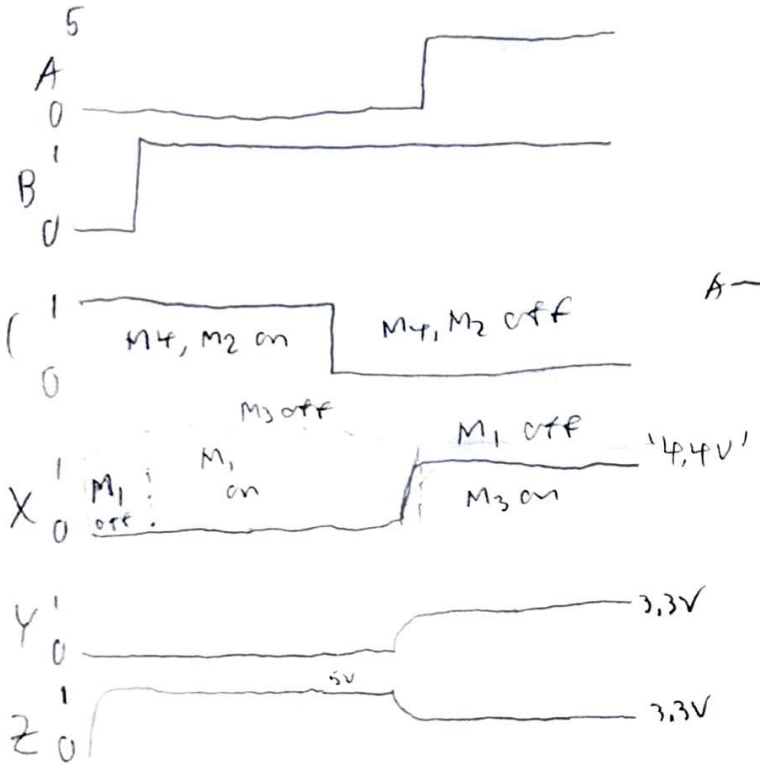
ECEN 153 HW7

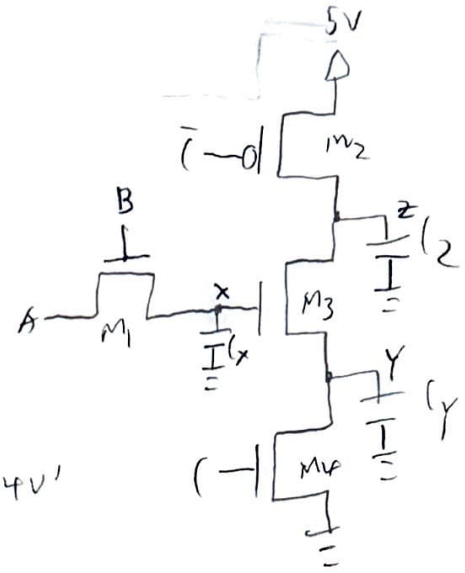
Dylan Thomas

$K_n = 115\ \mu A/V^2$ $V_{TN} = .43$ $V_{DSat} = .63$ $\lambda = 0$ $V_{dd} = 2.5$ $V_{Tbody} = .6$

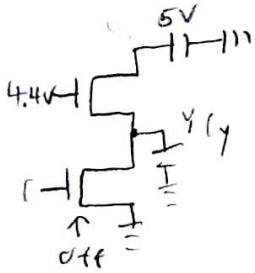
$K_p = 30\ \mu A/V^2$ $V_{Tp} = -.4$ $V_{DSat} = -1$ $\lambda = 0$ $V_{Tbody} = -.55$

$C_x = 20fF$, $C_y = 50fF$, $C_z = 100fF$





at x switching point



$5(100fF) + 0 =$

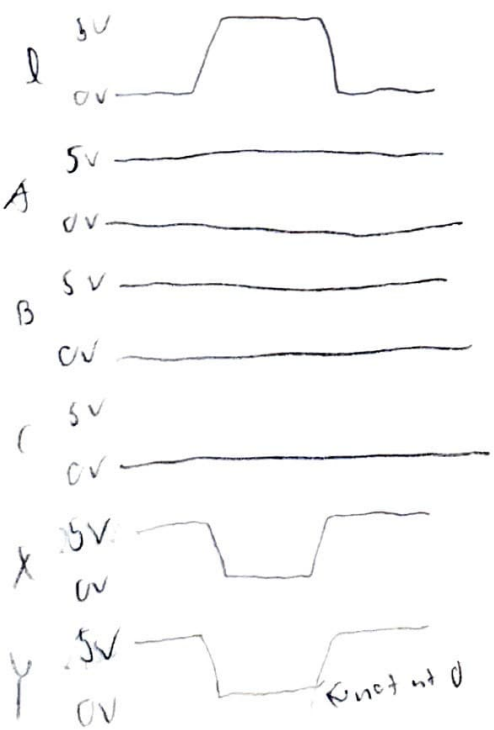
$V_{new}(100fF) + 3.8(50fF)$

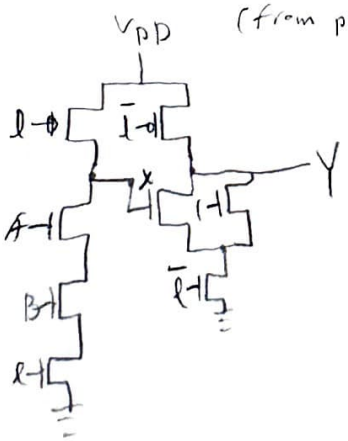
$V_{TPW} = 3.1$, will go

till V is equiv

$V_{final} = 3.\bar{3}V$

2.



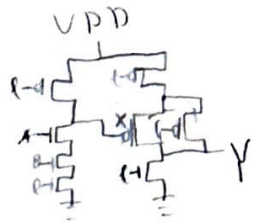


(from problem statement, $V_{DD} = 2.5V$

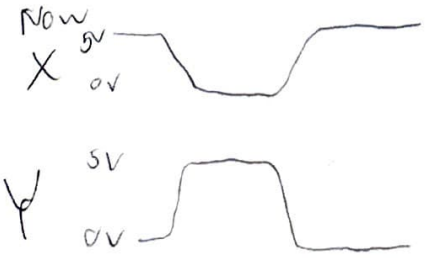
$X = \overline{ab}$

$Y = \overline{X + C} = \overline{\overline{ab} + C} = ab\bar{C}$

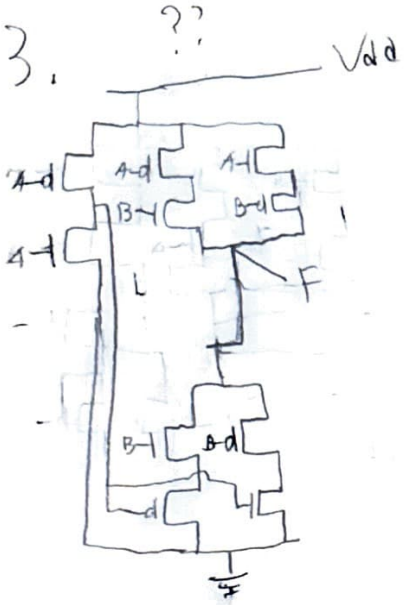
b. Use Nora Logic



Now



3. ??



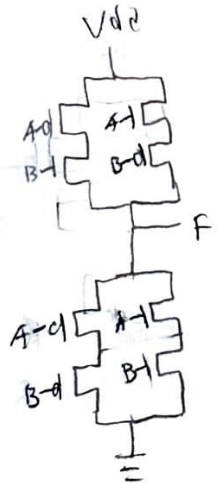
A	B	F	V
0	0	0	.55V
0	1	1	4.4V
1	0	1	4.4V
1	1	0	.55V

b. $F = A \oplus B$

c. bad XOR Gate (CMOS)

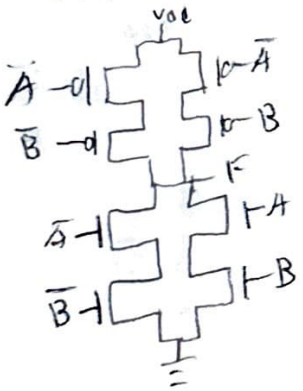
d. 3.85 V swing (4.4 − .55 = 3.85)

e. If we want to increase V swing, make a better CMOS XOR. However, this will use more transistors. To use less mosfets, we can do this, but that gives us an uneven V output



A	B	F	V
0	0	0	.55V
0	0	1	4.4V
1	0	1	4.4V
1	1	0	0V

The "better" CMOS XOR



Getting these 'nots' requires 2 more mos fets each (8 total).

8 + 8 = 16 total.